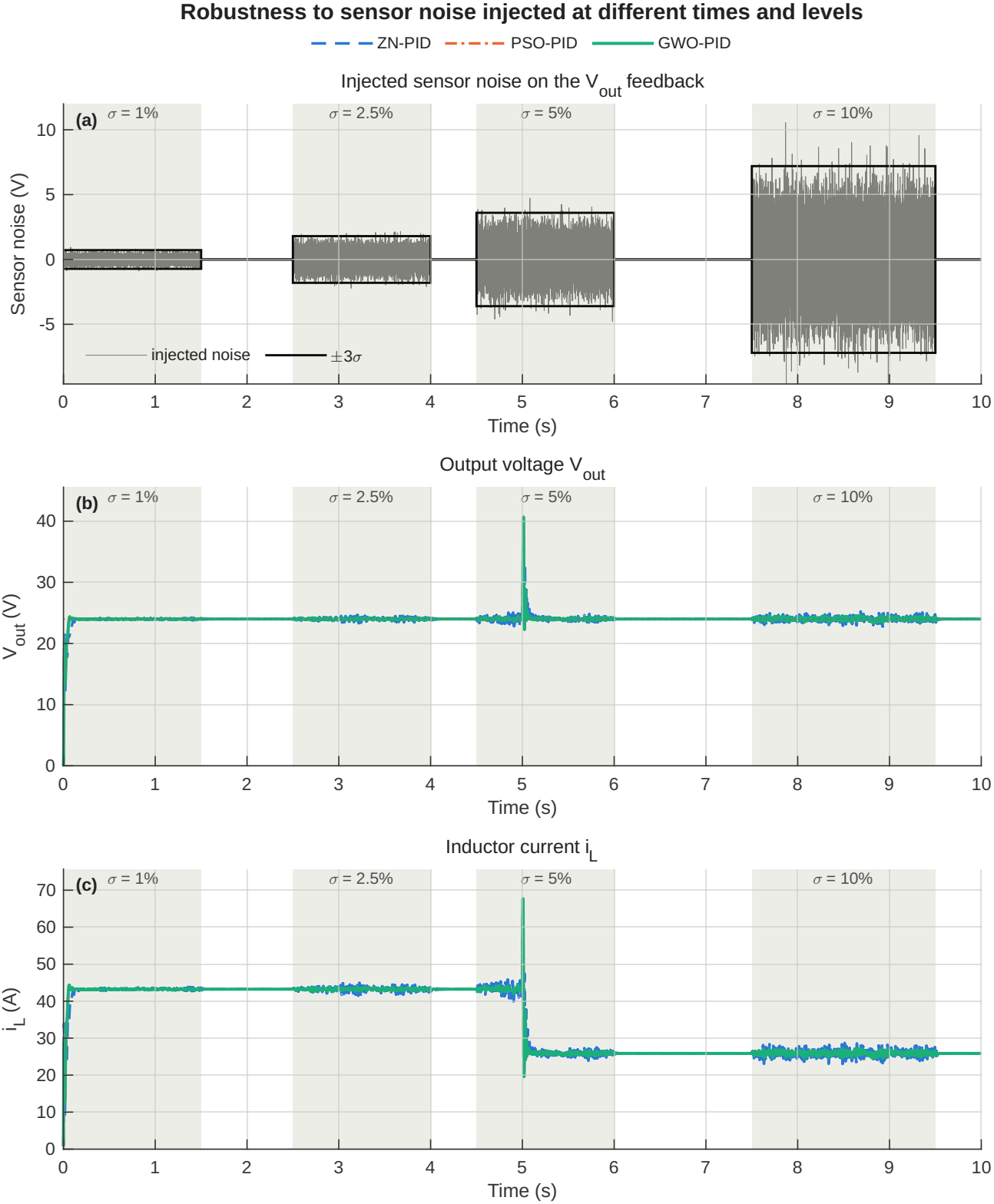

Robustness to sensor noise injected at different times and levels
ZN-PID
PSO-PID
GWO-PID
Injected sensor noise on the V_out feedback
(a)
σ = 1%
σ = 2.5%
σ = 5%
σ = 10%
Sensor noise (V)
injected noise
±3σ
Time (s)
Output voltage V_out
(b)
V_out (V)
Time (s)
Inductor current i_L
(c)
i_L (A)
Time (s)